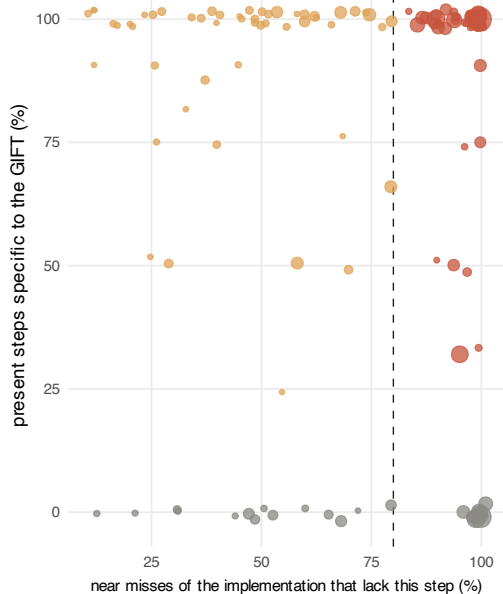


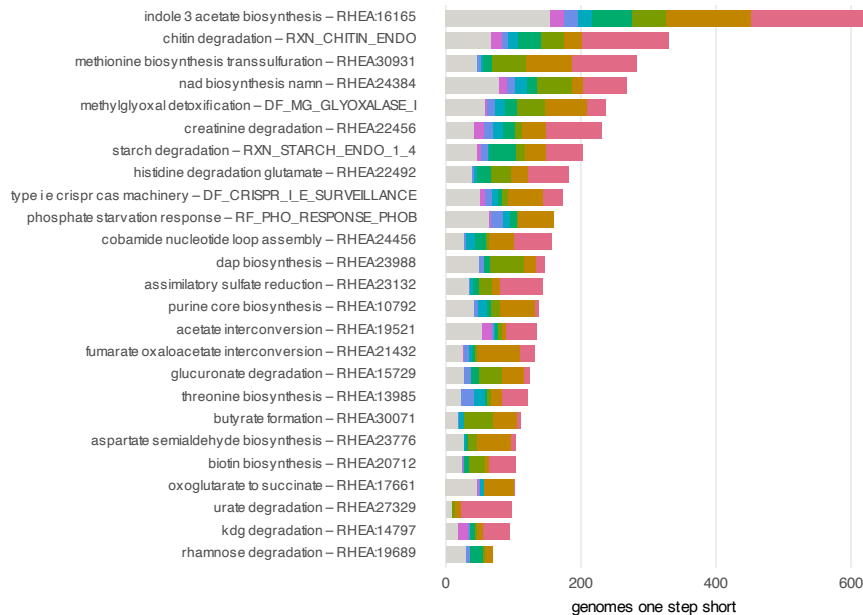
# Which step is missing when a GIFT is one step short

Each point is one implementation and missing requirement with at least 20 near-miss genomes. A present step is specific when no other GIFT requires it or accepts one of its markers. Right of the dashed line, at least 80% of the implementation's near misses lack the same step.

**a** Is it the same step, and are the present steps specific?



**b** Most frequent candidates, by phylum



● same step missing, specific steps present    ● missing step varies, specific steps present    ● present steps all shared with other GIFTs

genomes one step short    ● 50    ● 200    ● 600

■ Pseudomonadota    ■ Bacillota    ■ Bacteroidota    ■ Actinomycetota    ■ Desulfobacterota    ■ Planctomycetota    ■ Cyanobacteriota    ■ other phyla

Candidates (b) are the red points of a: the same step missing while GIFT-specific steps are present. A candidate is a prompt for curation: an uncured alternative, a marker that fails in a lineage, or a capability that is not encoded. A near miss is never counted as support.